

Manu Jhawar

B.S. Computer Science & Physics, University of California, Santa Cruz (Expected 2029)

mjhawar@ucsc.edu | github.com/manwhar | manwhar.net

EDUCATION

University of California, Santa Cruz

B.S. Computer Science & Physics — Expected 2029

GPA: 3.9 / 4.0

Relevant Coursework

Mathematics: Linear Algebra, Calculus II, Vector Calculus (*in progress*)

Computer Science: Introduction to Computer Systems (*in progress*), Introduction to Machine Learning (*in progress*), C Programming (*in progress*)

Physics: Classical Mechanics (*in progress*), Waves & Optics (*in progress*)

Additional Coursework (Foothill College)

Object-Oriented Programming (Python), Data Structures & Algorithms, Intermediate Software Design

TECHNICAL SKILLS

Tools: Python: NumPy, PyTorch, Matplotlib, Pandas; Git, Linux, Premiere Pro, Photoshop

Content Creation and Video Editing

2021 - Present

- 200+ videos (700k+ views): scripting, editing, audio/color; audience-focused storytelling
- Freelance video editor, including legal media

HONORS

UC Santa Cruz Dean's Honors

Foothill College Dean's List (3x)

California Seal of Biliteracy (Spanish)

RESEARCH & TECHNICAL PROJECTS

Interactive simulations and detailed lab reports available at manwhar.net

Pairwise Preference Study (15,000+ participants) ↗

- Large-scale study using pairwise comparisons to classify preferences of an online community
- Analyzed 417k+ crowdsourced datapoints, led community outreach, engineered data validation pipelines

Probabilistic Simulation & Analysis Projects ↗

- Developed simulation tools to model complex stochastic systems and analyzed emergent behavior
- Validated theoretical expectations against empirical results

Machine Learning Agent for Strategic Gameplay ↗

- Created PyTorch machine learning agents for decision-making in a stochastic card game environment
- Built state representations, reward shaping dynamics, and optimization strategies
- Gained practical experience with tensors, gradient-based learning, and model evaluation

Planetary, Particle, and Dust Simulations ↗

- Modeled celestial dynamics, subatomic particle interactions, and dust systems from scratch
- Built simulations to observe emergent wave phenomena, predict eclipses, and validated against empirical formulae

EMPLOYMENT & LEADERSHIP

Barista, Peet's Coffee & Tea

Aug 2024 - Aug 2025

Worked 20 hrs/week alongside full academic course load.

Student Government and Media Leadership (HS)

Aug 2023 - Jun 2025

Class President, ASB Secretary, Video Announcements Lead